

Event : $\{t, x, y, z\}$

Instantaneous : $\delta(t)$

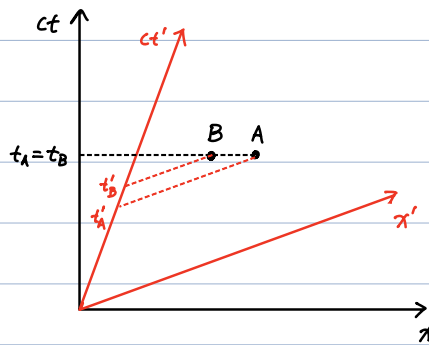
point-like : $\delta(x)\delta(y)\delta(z)$

Simultaneous events : $t_1 = t_2$ (which frame?)

Q: In frame K, two events happen at the same time; In K' frame, still same time?

• Newtonian physics : Yes

• Relativity : NOT always.

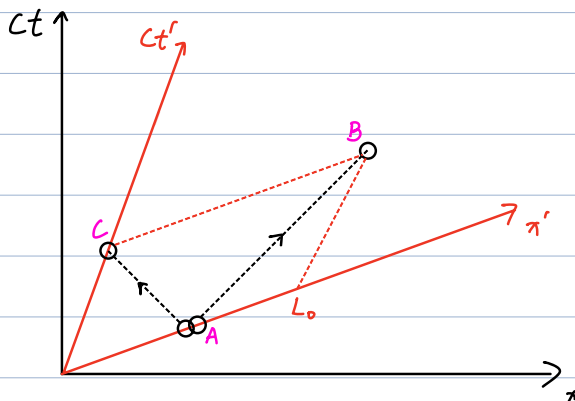
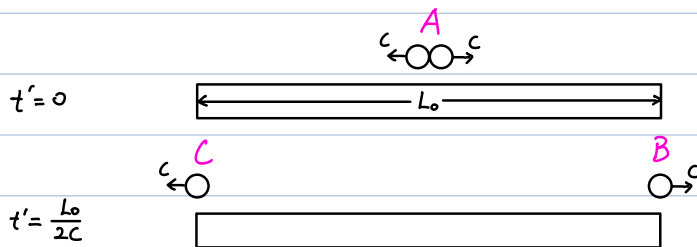


Simultaneous events A & B in frame K

$$\begin{cases} t_A = t_B \\ t'_A \neq t'_B \end{cases}$$

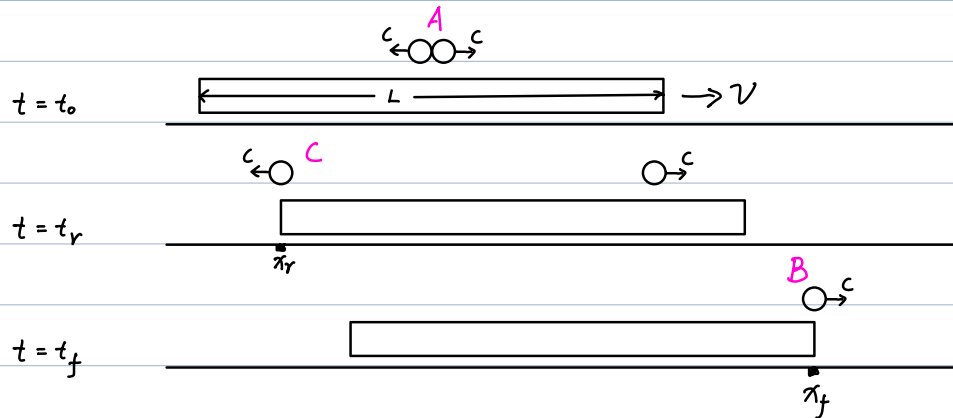
Example.

In a train frame (K')



keep

In the track frame (K)



B & C NOT simultaneous in K-frame!

Denote $\begin{cases} T \equiv t_f - t_r \\ D \equiv x_f - x_r \end{cases}$

$$\begin{cases} c(t_r - t_0) = \frac{1}{2}L - v(t_r - t_0) \\ c(t_f - t_0) = \frac{1}{2}L + v(t_f - t_0) \end{cases}$$

$$\Rightarrow cT = v(t_r + t_f - 2t_0)$$

Also, since

$$c(t_r - t_0) + c(t_f - t_0) = D$$

$$\Rightarrow \boxed{T = \frac{v}{c^2} D}$$

time difference in K of 2 simultaneous events in K' $\swarrow$ spatial difference in K of 2 simultaneous events in K'

• Method-2.

Simultaneous events in K' at time

$$\text{B} \begin{cases} t' = \frac{L_0}{2c} \\ x'_f = L_0 \end{cases} \quad \text{C} \begin{cases} t' = \frac{L_0}{2c} \\ x'_r = 0 \end{cases}$$

Lorentz TR:

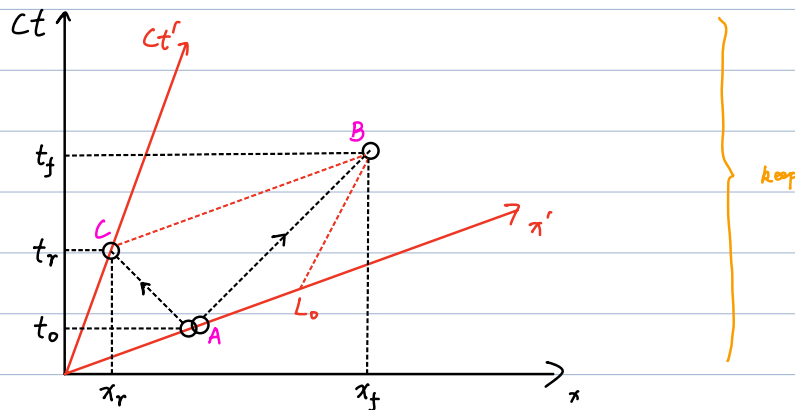
$$\text{C:} \quad \begin{pmatrix} ct_r \\ x_r \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \frac{L_0}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}\gamma L_0 \\ \frac{\beta}{2}\gamma L_0 \end{pmatrix}$$

$$B: \begin{pmatrix} ct_f \\ x_f \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \frac{L_0}{2} \\ L_0 \end{pmatrix} = \begin{pmatrix} \gamma L_0 (\beta + \frac{1}{2}) \\ \gamma L_0 (\frac{\beta}{2} + 1) \end{pmatrix}$$

$$\Rightarrow \begin{cases} cT = \beta\gamma L_0 \\ D = \gamma L_0 \end{cases}$$

$$\Rightarrow T = \frac{v}{c^2} D$$

- Viewing from Minkowski diagram (Method-3)



$$\begin{cases} \tan \theta = \frac{c(t_f - t_r)}{x_f - x_r} \\ \tan \theta = \frac{v}{c} \end{cases}$$

$$\Rightarrow T = \frac{v}{c^2} D$$

Q: Still holds if replacing photons by balls with speed u (in K')?

- Method-2:

Simultaneous events in K' :

$$B \begin{cases} t' = \frac{L_0}{2u} \\ x'_f = L_0 \end{cases} \quad C \begin{cases} t' = \frac{L_0}{2u} \\ x'_r = 0 \end{cases}$$

Lorentz TR:

$$C: \begin{pmatrix} ct_r \\ x_r \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \frac{cL_0}{2u} \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma \frac{cL_0}{2u} \\ \beta\gamma \frac{cL_0}{2u} \end{pmatrix}$$

$$B: \begin{pmatrix} ct_f \\ x_f \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \frac{cL_0}{2u} \\ L_0 \end{pmatrix} = \begin{pmatrix} \gamma \frac{cL_0}{2u} + \beta\gamma L_0 \\ \beta\gamma \frac{cL_0}{2u} + \gamma L_0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} cT = \beta\gamma L_0 \\ D = \gamma L_0 \end{cases} \Rightarrow \frac{cT}{D} = \frac{v}{c} \Rightarrow T = \frac{v}{c^2} D \quad \text{SAME as before}$$

- In fact, from Minkowski diagram (Method-3):

$$\frac{cT}{D} = \tan \theta = \frac{v}{c} \quad \text{still holds}$$

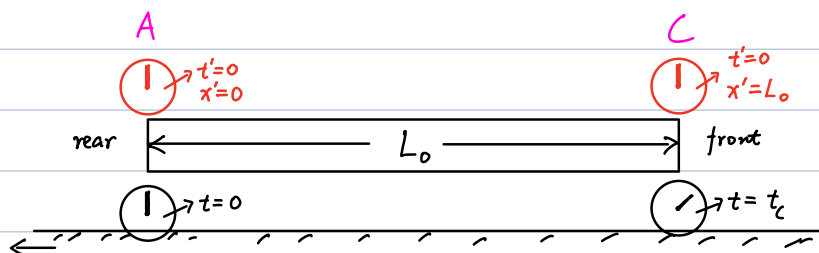
$$\Rightarrow T = \frac{v}{c^2} D$$

Length contraction & Time dilation

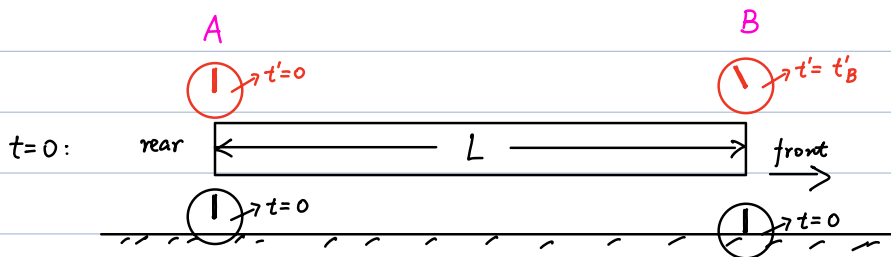
Proper length (L_0): length of object at rest (本征长度)

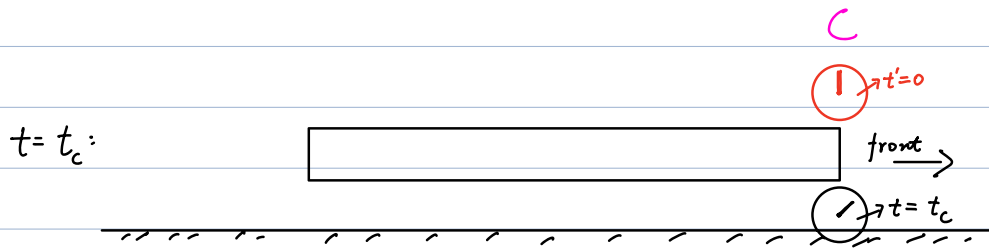
Example: $\xrightarrow{K'}$ train (moving to the right with v) & track $\xrightarrow{K}$

① train frame (track moves to the left)



② track frame



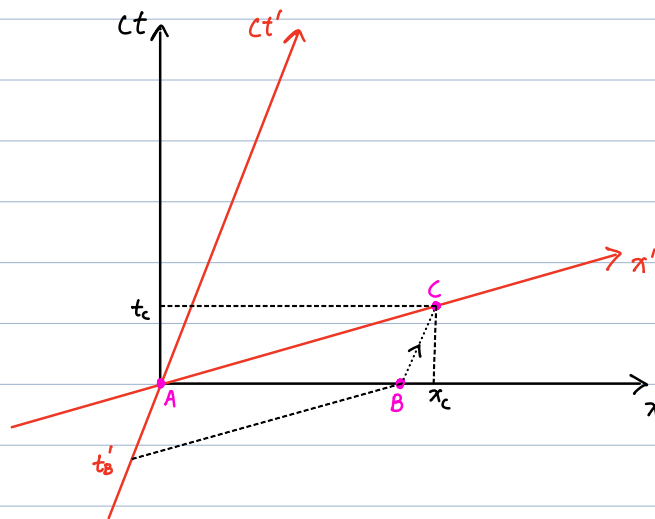


Events:

A: rear clocks meet at $t' = 0$ ($x' = 0$)

B: front clocks meet at $t = 0$ ($x = L$)

C: front clocks meet at $t' = 0$ ($x' = L_0$)



B $\rightarrow$ C (speed v):

$$v = \frac{x_c - x_B}{t_c - t_B}$$

Event C:

$$\begin{pmatrix} ct_c \\ x_c \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} 0 \\ L_0 \end{pmatrix} = \begin{pmatrix} \beta\gamma L_0 \\ \gamma L_0 \end{pmatrix}$$

$$\Rightarrow v = \frac{\gamma L_0 - L}{\beta\gamma L_0/c - 0}$$

$$\Rightarrow \beta^2 \gamma L_0 = \gamma L_0 - L$$

$$\Rightarrow \boxed{L = \gamma^{-1} L_0}$$

Comment: $\gamma^{-1} = \sqrt{1 - \beta^2} < 1$

$\Rightarrow L < L_0$

Length contraction: Length of moving object contracted!

B $\rightarrow$ C :

clock in K' : $t' = t'_B \rightarrow t' = 0$ $\Delta t_0 \equiv 0 - t'_B$

clock in K : $t = 0 \rightarrow t = t_C$ $\Delta t \equiv t_C - 0$

$$\begin{pmatrix} c t'_B \\ x'_B \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} 0 \\ L \end{pmatrix} = \begin{pmatrix} -\beta\gamma L \\ \gamma L \end{pmatrix}$$

$\Rightarrow \Delta t_0 = \beta\gamma L / c$

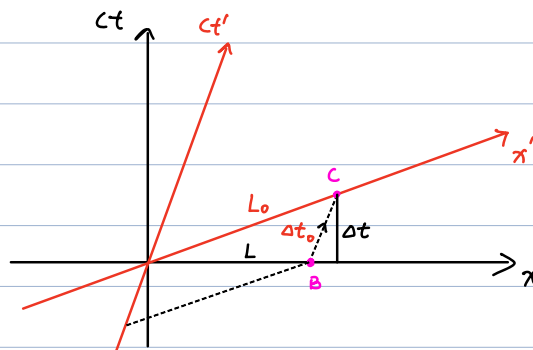
Also, $t_C = \beta\gamma L_0 / c$ (just derived)

$\Rightarrow \Delta t = \beta\gamma L_0 / c = \frac{L_0}{L} \Delta t_0$

$\Rightarrow \boxed{\Delta t = \gamma \Delta t_0}$ $\gamma > 1$

Time dilation: moving clock slows down

Geometrical meaning of "length contraction" & "time dilation":



$L = \gamma^{-1} L_0 < L_0$

$\Delta t = \gamma \Delta t_0 > \Delta t_0$